

DATA ANALYSIS REPORT

IT Support Cost Optimization

Beginner Level Internship Assignment - Data Analyst

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GitHub Link	https://github.com/SahilGurjarr/data-analyst-assignment

1. Introduction

I have worked on a small IT support ticket dataset to understand cost and ticket handling issue. The main aim is to understand where support cost is increasing and how ticket handling can be improved.

For this analysis, I used a small sample dataset with ticket category, priority, resolution time, cost, client type, escalation count and customer rating. I then made simple charts to find common patterns.

2. Problem Statement

The company handles many IT support tickets every year. Management wants to reduce support cost, improve customer satisfaction, reduce unnecessary escalations and improve engineer allocation.

3. Dataset Used

The dataset used in this report contains 50 sample IT support tickets. The table below shows the main columns used for analysis.

Column	Meaning
Ticket_ID	Unique ID of each support ticket
Issue_Category	Type of issue such as Network, Login or Software
Priority	Ticket priority: Low, Medium or High
Resolution_Time_Hours	Time taken to solve the ticket
Total_Cost	Estimated cost for resolving the ticket
Client_Type	Small, Medium or Enterprise customer
Escalation_Count	How many times the ticket was escalated
Customer_Rating	Customer satisfaction rating

4. Dataset Preview Screenshot

This screenshot shows a small preview of the ticket dataset used for analysis.

Ticket_ID	Issue_Category	Priority	Resolution_Time_Hours	Cost_Per_Hour	Total_Cost	Client_Type	Escalation_Count	Customer_Rating
T-1001	Software	Low	4	133	532	Small	3	4.1
T-1002	Software	Medium	3	147	441	Small	1	3.9
T-1003	Network	Low	8	178	1424	Enterprise	3	3.3
T-1004	Hardware	Low	4	135	540	Enterprise	0	3.9
T-1005	Hardware	Low	3	133	399	Medium	0	4.3
T-1006	Software	High	8	166	1328	Medium	3	3.2
T-1007	Network	Medium	5	179	895	Enterprise	2	3.5
T-1008	Software	High	5	154	770	Small	3	4.2
T-1009	Software	Medium	4	140	560	Small	2	3.7
T-1010	Security	Medium	6	167	1002	Medium	2	3.8
T-1011	Hardware	Medium	6	140	840	Small	0	4.2
T-1012	Login	High	2	95	190	Medium	1	4.7
T-1013	Login	Medium	1	112	112	Medium	1	4.3
T-1014	Network	Low	8	199	1592	Medium	2	3.3
T-1015	Software	Low	3	188	564	Medium	2	4.1
T-1016	Login	Low	1	112	112	Small	0	4.6
T-1017	Network	Medium	9	175	1575	Enterprise	2	3.1
T-1018	Network	High	7	191	1337	Enterprise	3	3.2
T-1019	Security	High	6	167	1002	Enterprise	3	3.9
T-1020	Security	Low	4	176	704	Medium	4	4
T-1021	Network	Medium	6	219	1314	Medium	3	3.8
T-1022	Hardware	Medium	4	156	624	Enterprise	2	4.1
T-1023	Login	Low	3	117	351	Small	0	4.2
T-1024	Hardware	Low	1	109	109	Small	2	3.9
T-1025	Software	Low	6	151	906	Small	2	4.1
T-1026	Network	Low	7	209	1463	Medium	3	3.2
T-1027	Software	Medium	5	133	665	Small	1	3.6
T-1028	Software	Low	6	129	774	Small	2	4.1
T-1029	Security	Low	8	171	1368	Enterprise	3	3
T-1030	Software	Low	3	168	504	Medium	1	4.3
T-1031	Login	High	4	126	504	Medium	1	4.3
T-1032	Login	High	3	127	381	Medium	1	4.2
T-1033	Software	Medium	4	127	508	Enterprise	4	3.5
T-1034	Network	Medium	6	201	1206	Enterprise	2	4
T-1035	Software	Medium	3	173	519	Small	3	3.9
T-1036	Software	High	5	153	765	Small	2	3.5
T-1037	Software	Medium	3	121	363	Enterprise	3	4
T-1038	Software	Low	6	136	816	Medium	2	4.2
T-1039	Security	Medium	8	198	1584	Small	1	3.3
T-1040	Software	High	5	134	670	Small	1	4.1
T-1041	Network	Low	7	183	1281	Small	3	3.6
T-1042	Security	High	8	165	1320	Small	1	3
T-1043	Network	High	10	204	2040	Small	3	3
T-1044	Network	High	9	197	1773	Enterprise	4	3.3
T-1045	Login	Low	1	93	93	Medium	0	4.7
T-1046	Login	Medium	3	127	381	Enterprise	0	4.8
T-1047	Network	Low	5	153	765	Enterprise	2	3.9
T-1048	Login	High	3	97	291	Enterprise	0	4.8
T-1049	Network	High	6	208	1248	Enterprise	3	3.5
T-1050	Hardware	Low	2	142	284	Small	1	4.4

5. Basic Analysis

After checking the data , I noticed these basic points :

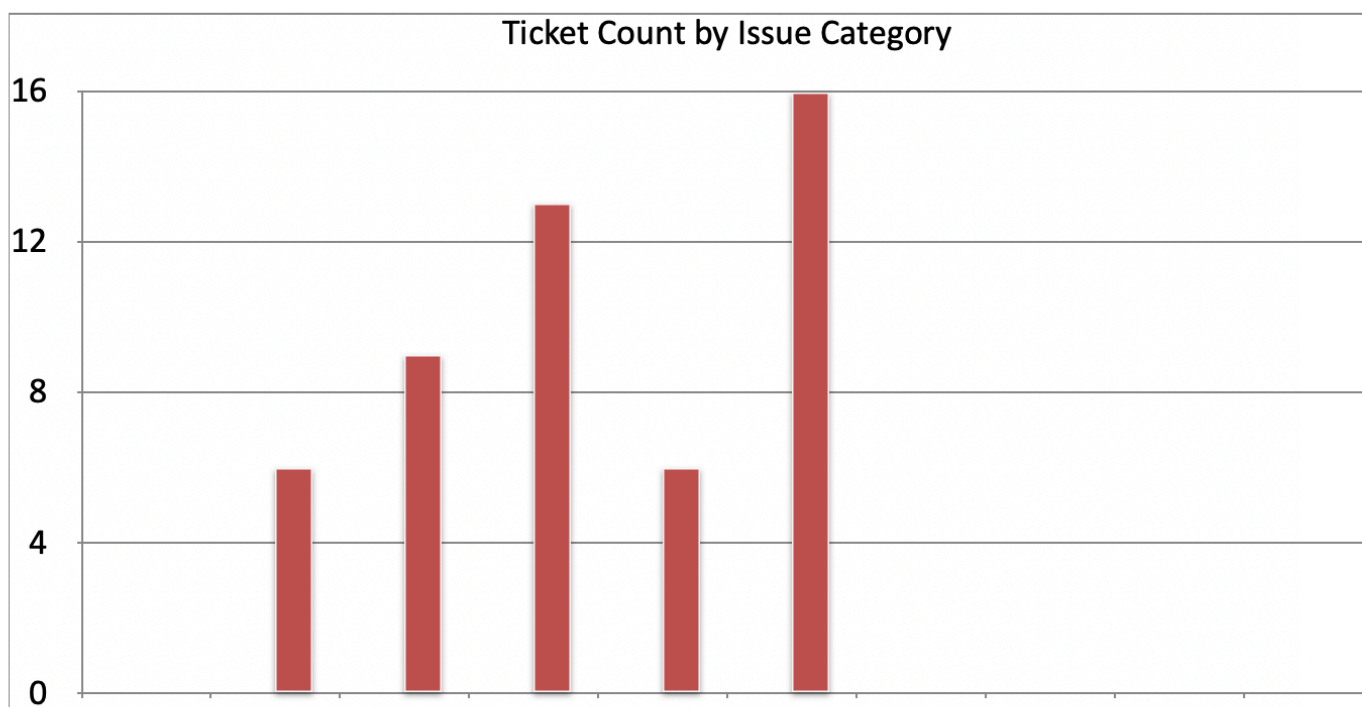
- Network and Security issues usually had higher cost.
- Login issues were solved faster and had lower cost.
- Enterprise clients had more escalations compared to small clients.
- High priority tickets generally took more time to resolve.

6. Output Screenshots - Charts

I created a basic a basic summary table and chart to understand ticket category count. And support pattern.

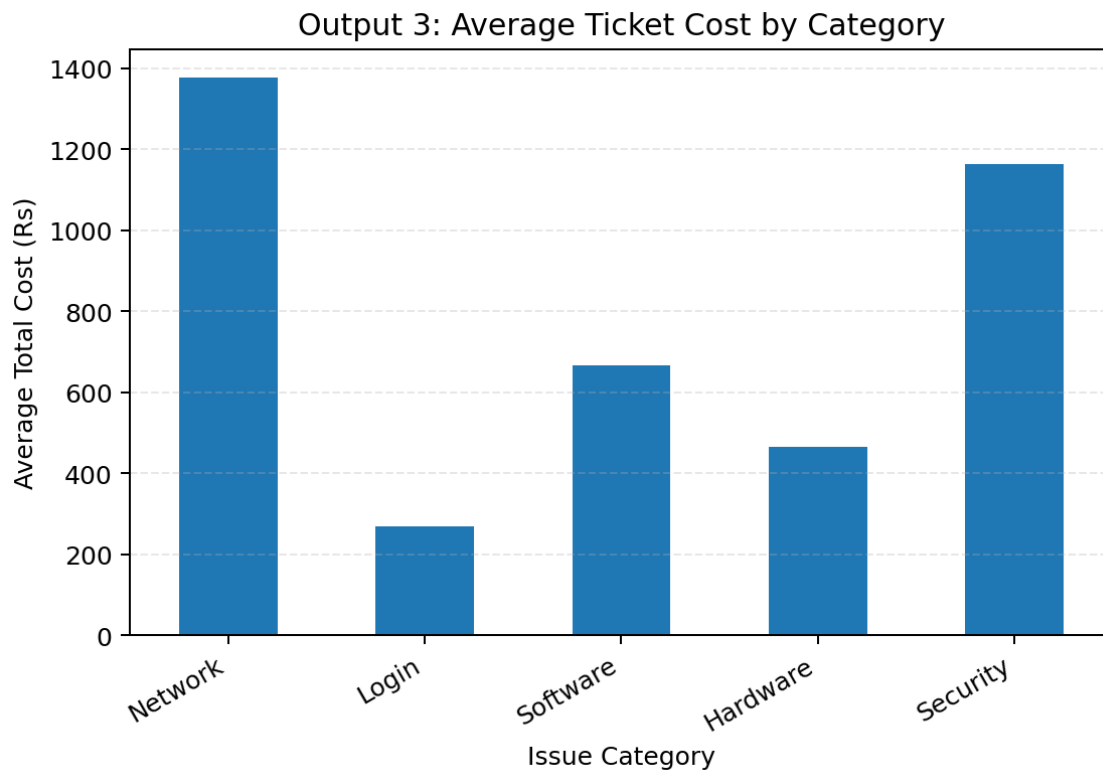
Chart 1: Ticket Count by Category

Issue_Category	Ticket_Count	Avg_Resolution_Time	Avg_Total_Cost	Avg_Escalation	Avg_Rating
Hardware	6	3.33	466	0.83	4.13
Login	9	2.33	268	0.44	4.51
Network	13	7.15	1378	2.69	3.44
Security	6	6.67	1163	2.33	3.5
Software	16	4.56	668	2.19	3.91



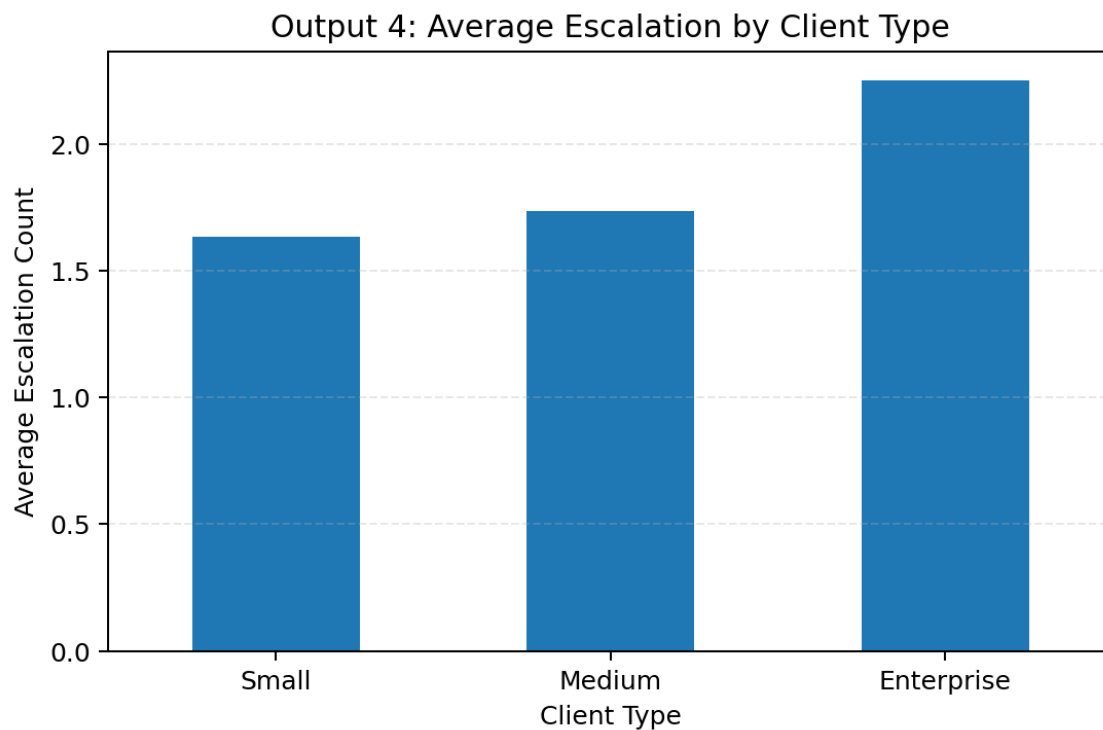
This chart shows which issue categories appeared more frequently in the dataset.

Chart 2: Average Cost by Category



This chart shows that some categories, especially Network and Security, had higher average ticket cost.

Chart 3 : Escalation Output



The chart shows that Enterprise clients had higher average escalation, mainly because their issues are usually more complex.

8. Recommendations

- Automate simple Login and Password related tickets to reduce engineer workload.
- Assign experienced engineers to Network, Security and other high priority tickets.
- Improve the ticket form so issues are categorized correctly at the start.
- Create faster support routing for Enterprise clients because they face more complex problems.
- Track support cost and escalation regularly using simple dashboards.

9. Conclusion

From this small analysis , I understood that simple charts can also help find useful business problems. By checking ticket cost, resolution time and escalation count, the company can identify where time and money are being wasted.

The main improvement areas are better ticket assignment, automation of simple issues, and faster support for high priority or Enterprise client tickets.